

Practice 6: Waveforming in Digital Modulation

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Abstract—In digital modulation, waveforming comprises the process by which a discrete sequence of modulated symbols is converted into a continuous waveform suitable for transmission. This process involves selecting a modulation scheme and then shaping each symbol in time and spectrum using pulse functions designed to control bandwidth occupancy and reduce inter-symbol interference (ISI). Techniques such as pulse shaping with Raised Cosine or Root Raised Cosine filters allow Nyquist criteria to be met and ensure efficient transmission without excessive overlap in the time or frequency domain. Waveforming also determines the envelope, spectrum, and behavior in the face of nonlinearities, directly influencing the signal-to-noise ratio (SNR), energy efficiency, and compatibility with power amplifiers. Overall, waveforming is an essential element in digital modulation, as it defines the final shape of the transmitted signal and conditions the spectral efficiency and reliability of the link.

keywords: *Inter-symbol interference, Modulation, Pulse function, Raised cosine, Root raised cosine, SNR, Efficiency.*

I INTRODUCTION

In digital communication systems, transmission quality critically depends on how the pulses carrying the information are shaped and filtered. The pulse shaping process aims to limit the occupied bandwidth, control intersymbol interference (ISI), and ensure that the final signal is compatible with the characteristics of the channel and the transmitter elements[1].

A useful starting point is the ideal rectangular filter, which conceptually allows each symbol to be transmitted without overlap within its time interval. However, although its temporal shape is simple, its spectrum exhibits sinc behavior with side lobes that extend indefinitely, implying inefficient use of the spectrum and sensitivity to bandwidth-limited channels. To improve this situation, filters specifically designed to meet the Nyquist criterion are used, among which the Raised Co-

sine (RC) filter and its Root Raised Cosine (RRC) version stand out. These filters allow spectral occupancy to be controlled by means of a roll-off parameter (β) and, at the same time, minimize ISI, ensuring that pulses are canceled at specific instants and do not interfere with neighboring symbols.

The behavior of these waveforms is analyzed both in the time domain, where pulse overlap and smoothing are observed, and in the frequency domain, where it is possible to compare the spectral efficiency and out-of-band attenuation of the different filters. A fundamental tool for evaluating the quality of the received signal is the eye diagram, which allows the visualization of the temporal window available for symbol decision and the detection of distortions such as jitter, ISI, and noise. A wide-open eye indicates a reliable link, while a closed eye indicates significant degradation, typically associated with inappropriate filtering, channel dispersion, or excessive ISI.

Identifying ISI is essential because it allows us to understand how the choice of transmission and reception filters, as well as the symbol rate and channel bandwidth, directly affect system performance. When pulse shaping does not satisfy the conditions of temporal orthogonality, neighboring symbols overlap and generate decision errors in the receiver.

The presence of noise in the channel, such as additive Gaussian white noise (AWGN), introduces additional degradation that manifests itself in the dispersion of the constellation points and in the closure of the eye diagram. The joint study of filtering, temporal/frequency analysis, ISI, and noise provides a comprehensive understanding of the actual behavior of a digital system and highlights the importance of correctly selecting filters and modulation parameters to achieve robust and efficient transmission.

II THEORETICAL FRAMEWORK

II-A Ideal Rectangular Filter

The ideal rectangular filter is the starting point in the study of pulse shaping. Its function is to transmit pulses that occupy exactly one symbol interval, without overlapping in time.

The time equation of the ideal rectangular pulse of duration T is expressed as:

$$p(t) = \begin{cases} 1, & 0 \leq t < T, \\ 0, & \text{in other case.} \end{cases} \quad (1)$$

Its Fourier transform is given by:

$$P(f) = T \text{sinc}(fT), \quad (2)$$

which shows that the spectrum extends infinitely due to the non-smooth nature of the pulse.

II-B Raised Cosine Filter

The Raised Cosine filter is designed to meet the Nyquist criterion and reduce intersymbol interference (ISI). It controls spectral occupancy using the roll-off parameter β .

The frequency response of the Raised Cosine filter is given by:

$$H_{RC}(f) = \begin{cases} T, & |f| \leq f_1, \\ \frac{T}{2} \left[1 + \cos \left(\frac{\pi T}{\beta} (|f| - f_1) \right) \right], & f_1 < |f| \leq f_2, \\ 0, & |f| > f_2. \end{cases} \quad (3)$$

$$f_1 = \frac{1 - \beta}{2T}, \quad f_2 = \frac{1 + \beta}{2T}.$$

Its corresponding time equation is:

$$p_{RC}(t) = \frac{\sin(\pi t/T)}{\pi t/T} \cdot \frac{\cos(\pi \beta t/T)}{1 - (2\beta t/T)^2}. \quad (4)$$

II-C Root Raised Cosine Filter

The RRC filter is typically used in transmission and reception (divided into two identical filters), so that the cascade of both produces the complete Raised Cosine and satisfies the Nyquist criterion.

Its frequency response is defined as the square root of the Raised Cosine:

$$H_{RRC}(f) = \sqrt{H_{RC}(f)}. \quad (5)$$

This filter guarantees temporal smoothness and better spectral behavior, making the resulting signal suitable for band-limited channels.

III METHODOLOGY

This section presents in an orderly and detailed manner the experiments carried out during the laboratory, including simulations and other considerations addressed within the practice.

III-A Flowchart for simulated tests

The diagram in Fig. 1 shows the set of sections that make up the flowchart used to carry out the exercise and their respective functions. It should be noted that three relevant changes were made for the experiments carried out. Initially, the value of the symbol rate R_s was changed to that proposed in the exercise, which in this case is $48k$ [Baud]. For experiments with infinite BW (without filtering), the low-pass filter implemented after the FIR interpolation filter was deactivated. Finally, for the implementation of the different filters worked on, a section of variables was identified that defined the functions of each one, so depending on the experiment, the window was activated to be used as a structure condition in the low-pass filter starting from h, where h is the vector that models this type of window. The value of h was standardized at 16 for all experiments performed in order to control the bandwidth and shape the pulse, avoiding ISI.

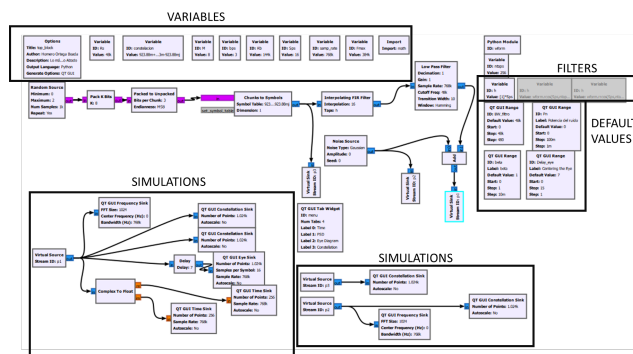


Figure 1. Initial schematic of the flowchart used to make the waveforming experiments.

The signal processing begins by generating a random signal with an amplitude of $2[V]$, which is then packaged into 8-bit words. Next, a packed-to-unpacked block is applied to generate n-bit packets defined by the modulation type based on the expression $\log_2(M)$, where M is the number of points used in the complex plane. Subsequently, chunk to symbol is used to map the packets in the complex plane. After this entire process, the respective filtering is performed and the simulations are obtained in time, frequency, eye diagram, and constellation[2].

III-B Parameter configuration

The configuration used for the experiments can be seen in TABLE 1. To complement this, the values used for the experiments without noise power are the same. The modulation implemented is 8PSK, and for the experiments with Gaussian noise, a P_n of 0.01[V] is used.

Without Noise ($P_n=0$)			
Case	BW	Beta	Filter
Case 1	∞ kHz	—	Ideal
Case 2	48 kHz	—	Ideal
Case 3	48 kHz	1	Raised Cosine
Case 4	48 kHz	0	Raised Cosine
Case 5	48 kHz	0.5	Raised Cosine
Case 6	48 kHz	0.5	Root Raised Cosine

Table 1

PRESELECTED CONFIGURATION PARAMETERS USED IN THE SETUP OF THE LABORATORY EXPERIMENTS.

IV EXPERIMENTS AND RESULTS

IV-A Analysis of the system without Gaussian noise

Initially, it's proposed studying the system response for the cases specified above. For this first part of the section, it's study the system without adding Gaussian noise. The results of the experiments are shown in Fig. 2,3 for ideal window, Fig. 4,5 for raised cosine filter and Fig. 6,7 for root raised cosine.

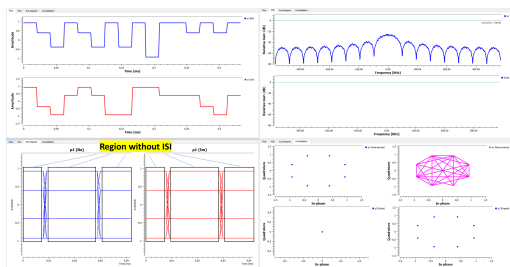


Figure 2. System output for case 1. Implementation without low-pass filter for 8PSK modulation. The graph shows the signaling of moments without ISI.

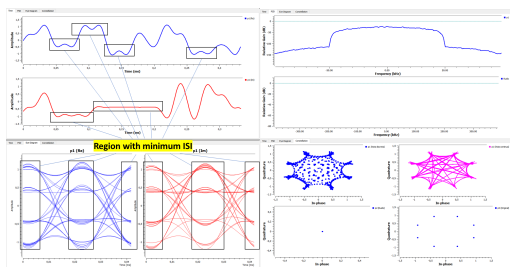


Figure 3. System output for case 2. Implementation with ideal low-pass filter using $BW = R_s$. The graph shows the signaling of moments with minimum ISI.

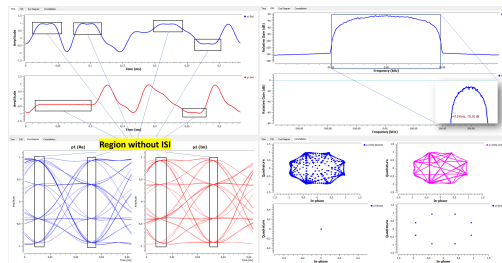


Figure 4. System output for case 3. Implementation with raised cosine filter for 8PSK modulation. The graph shows the signaling of moments without ISI. In addition, the bandwidth is shown in its PSD representation. The factor β takes the value 1.

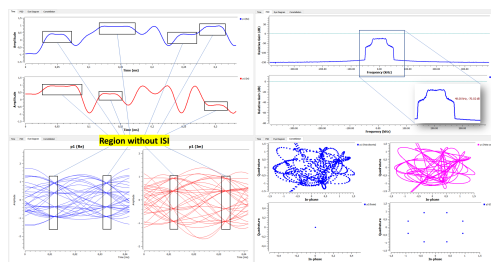


Figure 5. System output for case 4. Implementation with raised cosine filter using a β of 0. The graph shows the signaling of moments without ISI. In addition, the bandwidth is shown in its PSD representation.

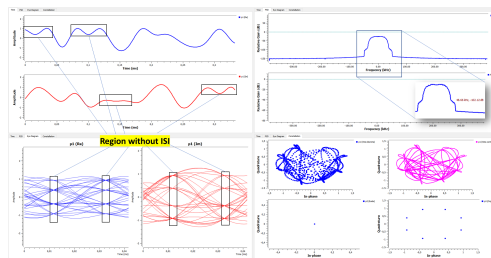


Figure 6. System output for case 5. Implementation with raised cosine filter using a β of 0.5. The graph shows the signaling of moments without ISI and BW.

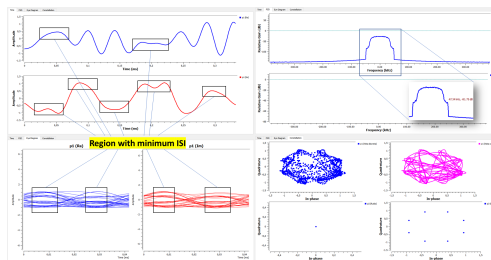


Figure 7. System output for case 6. Implementation with root raised cosine filter using the same β of experiment in Fig. 6. The graph shows the signaling of moments with minimum ISI and the BW obtained.

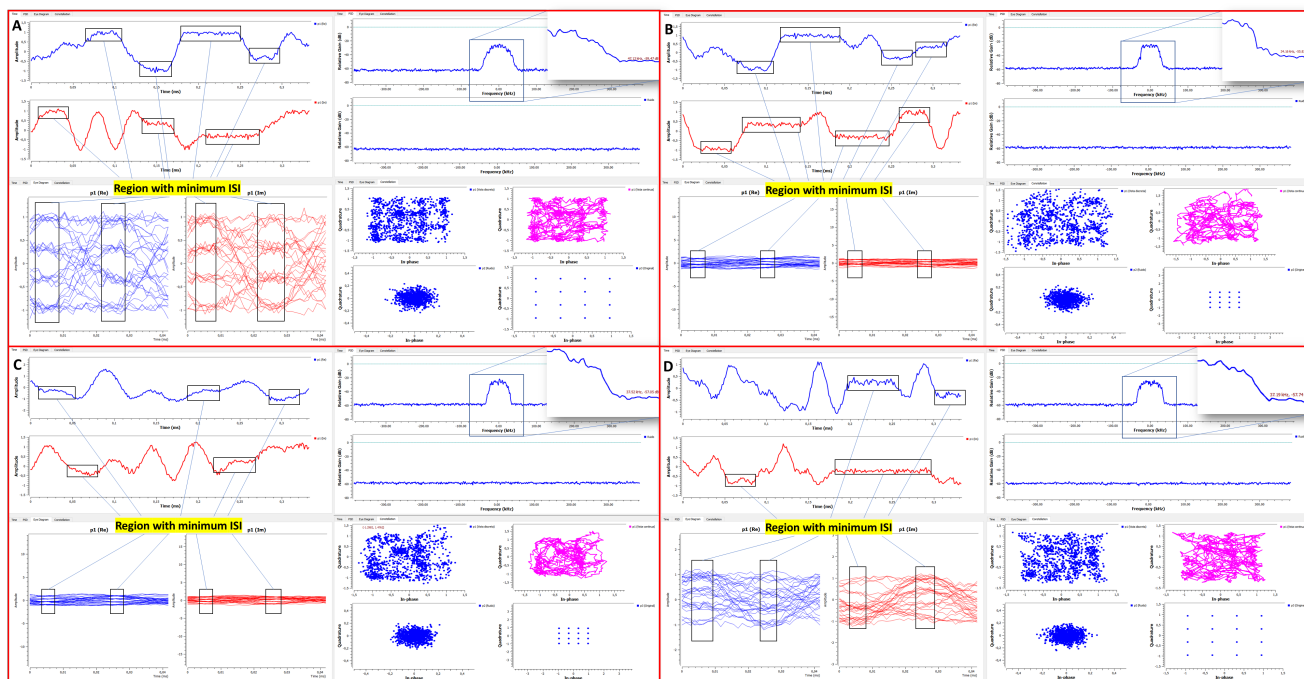


Figure 8. Diagram of condensed results from simulations with additive noise of 0.01[V], using 16QAM modulation. The figure is divided into four quadrants where the performance of the system is analyzed for the cases of cosine-raised filtering and root cosine-raised filtering previously discussed. Each quadrant has a section corresponding to the type of experiment performed. Section A is the experiment performed for case 3 with $\beta = 1$. B is the experiment performed by replacing β with 0. C corresponds to the experiment using $\beta = 0.5$. Finally, D is the experiment where the raised cosine root filter is implemented with $\beta = 0.5$.

IV-B Analysis of the system with Gaussian noise

As shown in Fig. 8, the same experiments as in the previous subsection are performed for 16QAM modulation, adding Gaussian noise with a magnitude of 0.01[V]. However, in order not to bias the analysis with noise, the same process is also repeated for 8PSK modulation in order to identify characteristics between the responses and advantages within the implementation of filtering, as well as the impact of the roll-off factor on filters such as raised cosine and root raised cosine. the results are seen in Fig. 11. In each case, a clear reduction in bandwidth was observed, unlike when working without added noise. This may be mainly due to the fact that the noise level is too high, so the noise floor in frequency reduces the usable bandwidth. Furthermore, it is clear that the roll-off factor plays a fundamental role in the filtering process, proportionally minimizing ISI.

The experiments carried out with an ideal filter can be seen in Fig. 9, 12. For these experiments, the guidelines were also followed, identifying periods with no or minimal ISI and the respective eye diagram and constellation analysis.

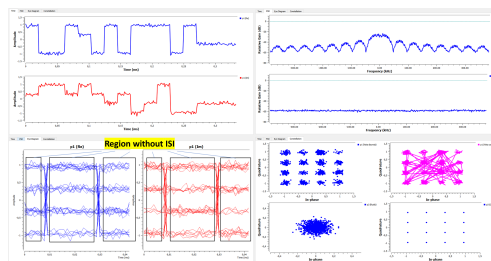


Figure 9. Flowchart output for 16QAM modulation with noise, without filtering.

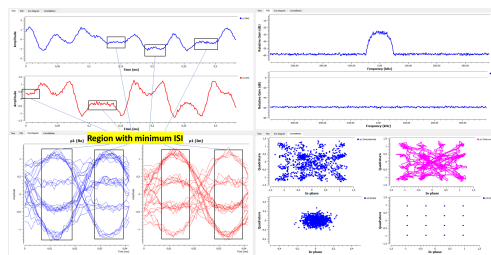


Figure 10. Case 2 with 16QAM modulation using an ideal filter with $BW = R_s$.

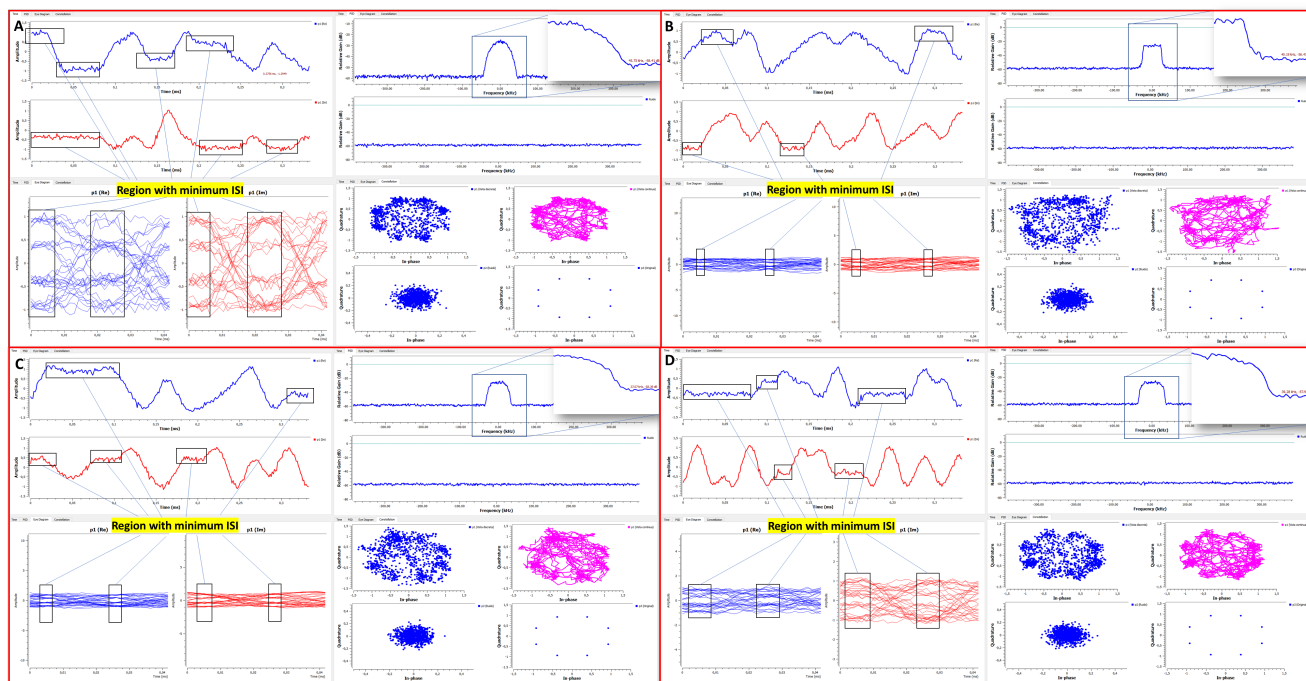


Figure 11. Diagram of condensed results from simulations with additive noise of 0.01[V], using 8PSK modulation. This figure follows the same distributions of the Fig. 8.

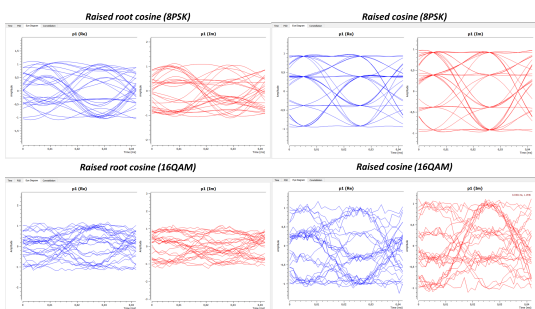


Figure 12. Comparison between eye diagrams obtained for a standard factor of $\beta = 0.5$ of cosine lift and cosine lift root.

The eye diagrams show a clear difference between using a raised root cosine filter alone and using a full Raised Cosine pulse shape. In both 8PSK and 16QAM, the raised cosine filter produces eyes that are more open and cleaner, indicating lower ISI and clearer decision regions. In contrast, the root raised cosine only cases exhibit greater trajectory spreading and partial eye closure, which is expected because this filter by itself does not satisfy the Nyquist criterion; it requires a matched root raised cosine filter at both the transmitter and receiver to form an equivalent RC response.

V CONCLUSIONS

The experiments confirm the fundamental role of pulse-shaping filters in controlling ISI in digital communication systems. The Ideal filter provides a useful theoretical reference but is not realizable in practice due to its infinite bandwidth. In contrast, the Raised Cosine filter demonstrates excellent ISI mitigation due to its compliance with the Nyquist criterion, resulting in clean, well-defined eye diagrams and efficient spectral confinement.

The root raised cosine filter, when used only at one side of the link, shows partial eye closure and increased symbol dispersion, highlighting that root raised cosine filtering must be applied symmetrically at both the transmitter and receiver to reproduce the raised cosine response and achieve optimal ISI suppression. This effect becomes more evident in higher-order modulation schemes such as 16QAM, where the eyes are more sensitive to distortions.

References

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